Semantically Enhanced IoT-Oriented Seismic Event Detection: An Application to Colima and Vesuvius Volcanoes

Mariarosaria Falanga, Enza De Lauro, Simona Petrosino, Diego Rincon-Yanez, and Sabrina Senatore, *Member, IEEE*

***Abstract*—Collecting massive seismic signals is a high-priority task in seismic risk evaluation, especially in densely populated areas, with cases of strong magnitude earthquake occurrence. At the same time, with the advent of the Internet of Things (IoT) paradigm, distributed and real-time environmental monitoring, supported by device interoperability, enhances the ability to collect data and make decisions especially in critical domains such as the seismic one. A crucial role is played by Semantic Web technologies that, in IoT ecosystems, promote syntactic and semantic interoperability, by enhancing the data quality that becomes ontology-annotated. This article introduces an IoT-oriented framework to collect seismic data, process and store them into a knowledge base. An ontology called Volcano Event Ontology (VEO) modeled for the seismic domain aims at gathering seismic signals collected by sensors for seismic event detection. The ontology is built on the well-known SSN/SOSA ontology, modeled to describe the systems of sensors, actuators, and observations. Seismic data have been collected by monitoring networks at Mt. Vesuvius (Naples, Italy) and Colima volcano (Mexico) and consolidated in the ontology. Moreover, the seismic data are also processed by a classification module to detect different seismic events (Volcano-Tectonic and long-period earthquakes, underwater explosions, and quarry blasts) and then stored in the knowledge base. Prompt detection and classification are, indeed, relevant to track any variation in the volcano dynamics, becoming crucial in cases of explosive crises. Finally, the VEO-driven knowledge base can be queried to get time-based seismic data and detected events, by queries.**

***Index Terms*—Internet of Things (IoT)-based sensor networks, machine learning, ontology, seismic domain, Vesuvius and Colima volcanoes.**

I. Introduction

INTERNET of Things (IoT) technology is becoming more appealing in recent years as an innovative option to environmental monitoring systems for many application domains: transportation, agriculture, healthcare, smart environment, and also natural disaster monitoring [1]–[3]. An IoT ecosystem consists of Web-enabled smart devices and sensors that interact and operate under a common format to collect, send, and act on data acquired from the environment. The high-level IoT applications aim at collecting data over time by the spatially distributed sensors and detect and recognize events in the real world to interpret the phenomenological data. In a geophysical context, in cases of natural disasters such as severe earthquakes or strong volcanic explosions, an accurate forecast of the occurrence and magnitude of both earthquakes and explosive phases is crucial for hazard assessment. For this purpose, to promptly identify anomalies, observatories worldwide manage different sensors collecting continuous seismic signals. The standard procedures for the seismic hazard infer the probability of future earthquakes from the statistical analysis of historical and instrumental earthquake catalogues [4]. Thanks to the broad deployment of the IoT-based system, real-time ground motion data will be available for monitoring and forecasting the activity of seismic events and also for assessing the structural dynamic behavior of the civil infrastructure [5]. However, the challenge around of data integration, exchange and interpretation and the IoT device interoperability, finds an effective solution in the use of Semantic Web Technologies (SWTs), for modeling and integrating data from heterogeneous and distributed sensor networks [6]. The synergy between IoT and SWTs promises to overcome the barrier of standardized heterogeneity and provides seamless solutions in remote sensing applications by taking benefits in knowledge representation, expression, and knowledge discovery.

The recent generation of sensing technology provides inherent support for processing massive data streams generated by sensors monitoring volcano activities. Besides the seismic networks with highly sensitive seismometers, data collection is recently supported also by IoT-based systems driven by the convergence of technologies such as microelectromechanical systems, wireless communication, and increased computing power. An example is MyShake [7], a smartphone application (from a collaborative project between UC San Diego and UC Berkeley) for measuring ground motion and detecting earthquakes from the background noise by aggregating multiple recordings from several smartphones. The IoT technologies are opening many avenues to enrich the sensing data

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sources, even in the seismic domain producing mobile, wearable, low-cost IoT sensors for earthquake detection. Consequent benefits are especially in developing countries, considering that the highly sensitive seismometers are expensive both for installing and maintaining them. In the light of these ongoing perspectives, this article presents an IoT-oriented framework to support the acquisition of seismic signals, processing, classification/recognition of the main seismic events. An ontology-based storage of the collected data allows retrieving time-based, aggregate information to support the humans in the scenario analysis and decision-making activities. The knowledge base is built on the ontology, Volcano Event Ontology (VEO), defined for the seismic domain, by exploiting raw data generated by IoT-based networks. Collected seismic data are relative to two very dangerous volcanoes, i.e., Vesuvius (Naples, Italy) and Colima (Mexico). VEO extends the well-known ontologies, namely, semantic sensor network (SSN) and sensor, observation, sample, and actuator (SOSA), designed to support the modeling of seismic sensors and stations.

In addition, the framework accomplishes seismic event detection by processing collected signals by machine learning techniques. To this purpose, a comparative analysis between multilayer perceptrons (MLPs) algorithm and convolutional neural networks (CNNs) has been carried out, aimed at assessing the classifiers performance in recognizing the seismic events.

This article is organized as follows. In Section II, related works are discussed, focusing the attention on volcano-seismology background as well as correlated artificial intelligence approaches. Section III introduces the seismological environment, providing details on the selected volcanoes, which represent the application domain. The ontology model and the machine learning approach are presented in Sections VI and VII, respectively, passing through Section IV, where the logical view of the proposed framework is given, and Section V in which the data acquisition module is explained. Then, Section VII reports the experiments that validate the effectiveness of the proposed classification models. Conclusions and future work follow in Section VIII.

II. Related Works

The spread of IoT technologies and the role of the Semantic Web in modeling sensor networks have highlighted new chances in the collection and integration of sensing data from different tools and heterogeneous sources. Especially in the Earth observations, sensor networks are low-cost solutions for monitoring land activities and mountain surveillance [8]. The role of SWTs in the sensor network domain provides many benefits to the data stream processing: for instance, the use of JavaScript Object Notation for Linked Data (JSON-LD, a method of encoding linked data based on the JSON format) represents an efficient way to semantically annotate data and, at the same time, collect an ontology-based description of the domain of interest to speed up the data filtering solutions.

SWTs allow data interoperability and integration, transforming low-level sensor data into high-level knowledge that becomes machine understandable; furthermore, they also provide reasoning capabilities that allow enriching the domain of interest with additional inferred information [9]. The semantics provides an underlying infrastructure for data representation, guaranteeing a shared agreement on the types of features, sources, and data. Besides these technologies aimed at structuring and managing the natural events and ecological activities, the role of ML is crucial to define accurate models for data analysis and classification tasks.

The remainder of this section provides an overview of the main works involving the research domains, such as semantic technologies and machine learning with reference to the sensing technologies domain; their synergy provides challenging solutions to face natural phenomena, specifically in the geological and seismological fields.

A. Ontologies and Sensing Technologies From the Seismological Viewpoint

With the IoT technological explosion and the need to extract data easily to interpret and process them, the semantic technologies represent an important framework, a step forward for collecting and processing semantically annotated data. The IoT technologies suffer from some drawbacks [3]: heterogeneity of physical objects, both regarding physical features and data and/or operations that they provide and/or accomplish; incomplete or missing metadata to describe things and their features. Moreover, the IoT end users are often not experts and they do not know IoT implementation details. These issues have been addressed by adding semantic annotations to the sensing devices: the ontologies have provided a seamless framework for formally describing the domain of interest. Semantic modeling produces an explicit description of the meaning of data in a structured way by merging domain knowledge and context-relevant information with raw measured data.

In the IoT domain, ontology modeling provides an explicit description of the things and devices, their features in a structured way by combining domain knowledge and contextual information with raw measured data, aimed at syntactic and semantic interoperability [10]. Several research works define high-level descriptions and semantic extension of *things* by means of ontology-based annotation. Just, for example, GoAAL [11] exploits SSN-based ontologies to support goal-oriented task in the ambient-assisted living (AAL) environment, specifically in the noncritical services for comfort and home management for elderly persons. In [12], a smart home ontology is proposed as a way to collect housing data and information about the appliances through the use of the non-IP mechanism with monitoring purposes.

The ontology-based development of IoT solutions is crucial to obtain universal solutions which ensure the adequate interoperability especially in a complex and risky environment.

The ontology modeling has been used indeed for describing natural phenomena or hazards [5], [13], [14]. In the seismic domain, volcano ontologies [15]–[17] are presented from different perspectives: for example, in [16], the eruptive styles and several dynamic volcanic processes that occur during eruptions are described from a physics viewpoint, whereas in [17], ontology modeling focuses on representing conceptual

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relationships between data and phenomena associated with volcanic domains, particularly linking plate tectonic features with volcanic systems. The proposed ontology VEO focuses mainly on the event identification by collecting sensing data available in the domain of interest. Like in [3], VEO uses SSN/SOSA ontologies to describe IoT sensors and sensing devices embedded on stations. The idea is to integrate into our knowledge base seismic data collected from traditional sensor-based station networks but also IoT-based sensor devices (e.g., wearable ones) to get a comprehensive and integrated view of what is happening from the seismic viewpoint. To support our proposal, recent literature proposes IoT-based systems based on a synergy of technologies, such as microelectromechanical systems, wireless communication, and increased computing power to collect seismic data [18]; some advances in mobile accelerometer technology reveal how smartphones [7] are becoming a valid alternative to fixed seismometers as the traditional sensing devices for the earthquake early warning [19]. Other alternatives to traditional networks include microelectromechanical system, accelerometers installed in buildings, USB accelerometers attached to personal computers, or other low-cost sensory equipment [20], [21].

The semantics provides a reliable infrastructure for data representation and integration. Semantic modeling of the seismic domain can describe a wide range of sensing data from heterogeneous sources: in addition to the traditional seismic network sensors also low-cost wearable sensors, are available with the advent of the IoT era. VEO provides a homogeneous data representation of different domains (seismic, sensing, geographical, etc.) by integrating heterogeneous data (IoT-based, seismologic, SAR, GPS, telemetric data, etc.) and ensuring syntactic and semantic interoperability. Behind the homogeneous sensing data representation, their storage in our knowledge base represents a common access point for the enhanced query-retrieved data view.

Other works in the literature use ontology-based technologies to model the seismic domain too, building information systems on seismology [22], also focusing on the earthquake emergency evaluation and response domain [5], [23].

B. Machine Learning in Volcano Seismology

In volcano seismology, several techniques of ML have been recently adopted, such as artificial neural networks (ANNs) and MLPs [1], [24], cluster analysis (CA) [25], self-organizing maps (SOMs) [26], support vector machines (SVMs) [27], CNNs [28], recurrent neural networks (RNNs) [29], and hidden Markov models (HMMs) [30].

For instance, with respect to Italian volcanic areas, the seismic signals were classified by using the classic ANN/MLP approaches, applied to Vesuvius [31], Stromboli [32], and Etna [33] mountains. Indeed, Scarpetta *et al.* [31] introduced an automatic classification of local seismic signals and Volcano-Tectonic (VT) earthquakes at Vesuvius based on MLP trained with a single hidden layer by using the quasi-Newton algorithm and the scaled conjugate gradient descent. Falsaperla *et al.* [32], by adopting the MLP approach combined with the back error propagation algorithm, concentrated the attention on the classification of the explosion quakes at the Stromboli volcano, identifying four different classes of shocks. Furthermore, Langer *et al.* [33] investigated the seismic tremor at Etna volcano by comparing the results achieved applying both supervised (SVM/MLP) and unsupervised (CA/SOM) techniques. CA algorithms are unsupervised networks very useful to look for similar shapes, independently of the sizes of the seismic events such as long-period (LP) signals that are generally clustered into families (see the case of Soufrière Hills Volcano, Montserrat [34]). In [35], the SVM method was used to detect seismic events by a distributed sensor network in the framework of the earthquake prediction in Portugal. RNN and CNN, widely employed in deep learning (DL) approaches, or ConvNets, which are a type of regularized MLP, are well-established methods in processing signals, specifically 1-D CNN [36]. Similar approaches was presented in [28] and [29] to classify different types of seismic events of the Colima volcano (Mexico). HMM classifiers have been adopted to classify the seismic events [30] in Colima and the Popocatepetl volcano (also located in Mexico). Recently, a deep neural network (DNN) with various architectures has been applied for automatic seismic event classification and detection. In this framework, it is noteworthy to cite the studies of Titos *et al.* [37] conducted at Colima to classify volcano-seismic events.

III. Seismological Background

In a volcano seismology framework, a great variety of waveforms can be recorded, such as VT earthquakes, LP and very LP (VLP) events, hybrids, low-frequency events (LFs), explosion-quakes and volcanic, and/or hydrothermal tremor. A prompt detection and classification of such signals is crucial for all the successive studies regarding their source mechanism and for monitoring purposes. Indeed, each event is generated by a different source mechanism, e.g., fracture dislocation; fluid/solid interaction; expansion of magma/gas bubbles; fluid-filled resonating cracks, or self-oscillating pipes (see [38], [39] and the references therein). Each source generally produces seismic signals in a specific frequency band but, often, they can be overlapped making the discrimination, based on the shape of the wave and on its frequency characteristics, very hard. Moreover, a standard practice of the operators working in the volcanology observatories is often to manually detect the events and this wastes much time, becoming unfeasible during a volcanic crisis. An automatic classification is thus relevant and crucial for a prompt detection of the volcanic source events especially during the nonstationary phase [1], [40], [41]. Additionally, active volcanoes require a permanent monitoring by means of spatially extended sensor networks, generating a big quantity of data whose streams have to be evaluated and processed. The enormous difficulties in manual/not-automated detection, the necessity to have a detection/classification as better as possible, and the request to deal with big data, have pushed researchers to make use of methodologies based on neural networks [42] up to the recent DL in order to make more accurate predictions On the other hand, the widespread of IoT technologies promises enhanced and low-cost sensor

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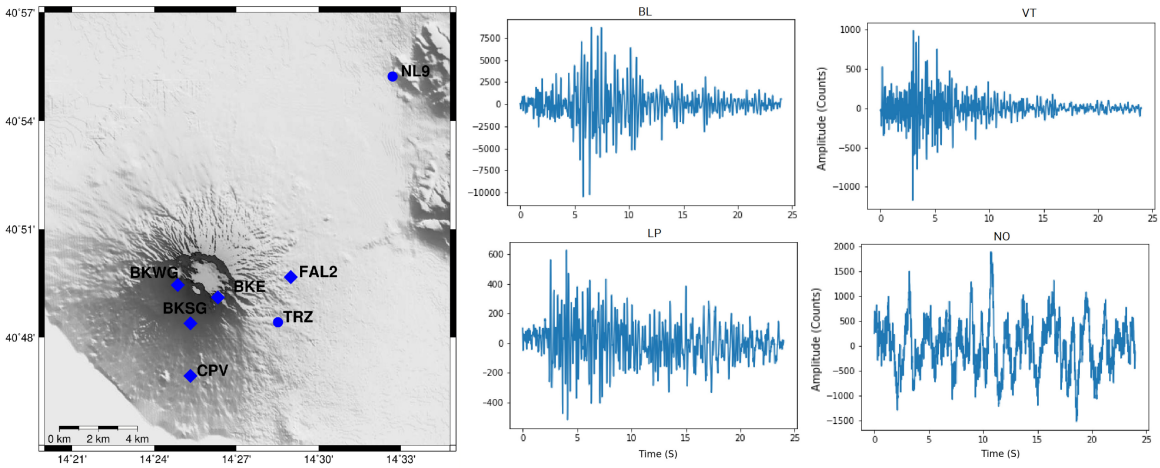


Fig. 1. Map of Vesuvius volcano location with the used three-component (blue diamonds) and one-component (blue dots) seismic stations of INGV-OV, on the left. An example of waveforms for each event category is reported on the right.

networks for efficient and accurate seismic signal acquisition and processing, along with a prompt communication among wireless-connected devices to propagate alerting situations, responsively.

A. *Vesuvius Volcano: Geological Environment*

The Mount Vesuvius is a very famous active stratovolcano known worldwide, located in the South of Italy, close to Naples (about 9 km). It is 1281-m high and 10-km wide and together with Somma volcano forms the so-called Somma-Vesuvius volcanic complex. The volcanism of that complex [43], [44] is essentially due to the compressive state of the Adriatic slab. The volcano has alternated Sub-Plinian and Plinian eruptions, like the one that destroyed Pompei town in 79 AD, together with much less intensity activity up to quiescent states along the hlcourse of millennia. The last eruption that occurred in 1944 had a violent Strombolian-like behavior. Currently, Vesuvius is basically in a quiescent phase with a few episodes of volcanic-hydrothermal nature [44]–[47]. Nevertheless, it is still considered one of the most dangerous and damaging volcanoes worldwide also because the metropolitan area, which extends up to its slopes, is characterized by the highest population density in Europe. The seismic activity recorded in the area mainly consists of VTs, with clear (primary) P and (secondary) S arrivals similar to pure tectonic earthquakes. They are generated by fractures that dislocate with shear failure at a depth of about 1–6 km beneath the volcanic edifice for the deep VTs, and at a few hundred meters a.s.l. for the very shallow VTs [44]–[46]. The magnitude of the shallow VTs is very low compared with the magnitude of the deep quakes that, in recent years, never passed the threshold of magnitude 3.6 relative to the earthquake that occurred on October 9, 1999 [46]. Recent investigations have also revealed some peculiar earthquakes with a lower frequency content compared with that of the VTs [44], [47], [48]. This kind of seismicity is atypical for the volcano and consists of LP and LF earthquakes. Both the event types are ascribable to the dynamics of the deep hydrothermal reservoir which induces variations of the fluid pore pressure in the medium [47].

Besides this seismic activity related to the dynamics at a regional scale or to a volcanic origin, other kinds of local signals can be observed at Vesuvius: man-made underwater explosions due to illegal explosive fishing, quarry blasts (BL), especially at Terzigno and Nola towns, and thunders (for a first classification see [31], [49]).

1) Vesuvius Sensor Network and Data Set: Analyzed data were recorded across the years 1999–2021 by the network of seismic stations managed by the Istituto Nazionale di Geofisica e Vulcanologia, sez. Napoli—Osservatorio Vesuviano (INGV-OV). Each station (which can be analog or digital) has a sensor/instrument, i.e., geophone or equivalently seismometer, and it is equipped with a data logger for local acquisition (for stand-alone installation) or with a transmission apparatus (to send real-time signals to the acquisition center). The seismic sensors are vertical single component or three components; depending on their natural frequency, they are short period or broadband instruments [31], [47]. Depending on the station, data are acquired with a sampling frequency equal to 100 or 125 Hz. The seismic stations used for the present analysis are reported on the map (see Fig. 1): BKE and BKWG are the closest to the crater, on the eastern and western slopes, respectively; BKSG is located south to the crater; TRZ and FAL2 are in the western sector, near the town of Terzigno; NL9 is located in Nola town; and CPV is sited on the coast of the Gulf of Naples.

Our dataset is composed of VTs, LPs, blasts, and seismic noise. The lowest magnitude VTs are recorded only by the summit station BKE; whereas the most energetic VTs are recorded by all the stations of the seismic network. LP events are less frequently recorded compared with the VTs; they are mainly detected by the stations close to the crater area (BKE, BKWG, and BKSG). Blasts have origin in both illegal fishing and cave extraction activity. Sea explosions are recorded by the CPV station, while quarry blasts are recorded by the closest TRZ and NL9 stations. Especially, the NL9 station records the seismic signals due to mining activity in the Nola cave. However, energetic explosions are also recorded at BKE.

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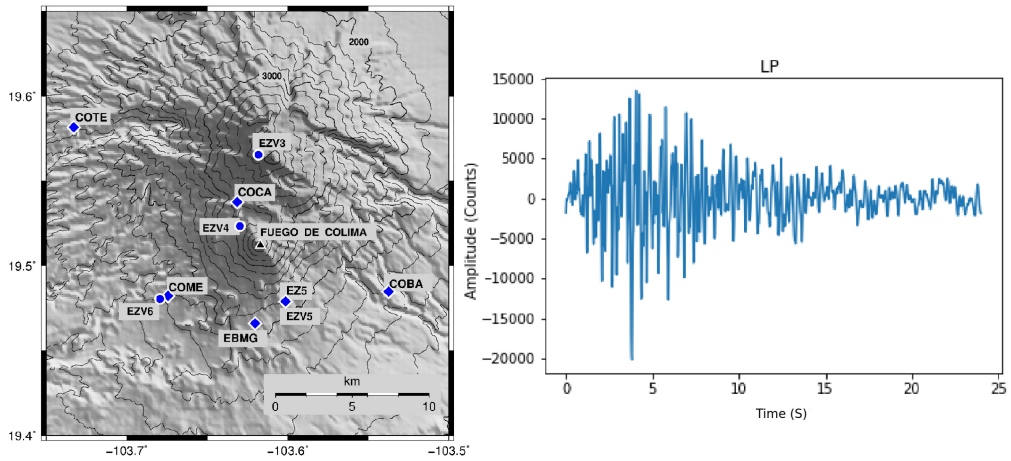


Fig. 2. Map of Colima volcano location with the used three-component (blue diamonds) and one-component (blue dots) seismic stations on the left [52]. An example of the LP event is reported on the right.

B. Colima Volcano: Geological Environment

El Volcán de Fuego de Colima [50], also known as Volcán de Colima (3860 m a.s.l.) together with the Nevado de Colima, forms the Colima Volcanic Complex (CVC). It is placed within the Colima graben at the western part of the Mexican Volcanic Belt and is a youthful andesitic stratovolcano within a 5-km-wide caldera. The Colima Volcano represents one of the most active and dangerous volcanoes in North America. Hence, the scientific monitoring of the volcano began in the middle of the 19th century and a variety of eruptive styles has been detected. Frequent historical big eruptions date back to the last 500 years and the last large explosive one occurred on January 20, 1913 lasting for only four days. In [51], a historical eruptive review can be found; recent volcanic activity is exhaustively described in [50].

The present activity of Colima is characterized by a variety of eruption styles, comprising dome growth, lava flows with numerous rockfalls, phreatic explosions, larger explosive events, and pyroclastic flows. Since November 1997, several eruptive cycles characterized by the growth of a lava dome accompanied by lava flows and small explosions, and the subsequent destruction of the dome due to a larger explosion have alternated [53]. In the period July–December 2005, the explosive activity was characterized by two until five explosions per day. The rate of the events occurrence increased slightly from January to March 2006, then it continued stable for the end of the year. Since January 2007, the number of small explosions grows allowing to develop a new lava dome in the crater in February 2007. The dome becomes higher than the crater rim, filling about 80% of the crater itself.

The recorded seismic signals display a wide range of waveforms consisting in VT and LP events, with the last one associated with source pressurization processes preceding Vulcanian-like explosions or ash-free events [54], [55]. All the seismicity at Colima volcano has been approximately located in the crater area at a few kilometers (less than 4 km) of depth [52], [56]–[58].

1) Colima Sensor Network and Data Set: We have studied the LP seismicity recorded in the period November 2005–April 2006 acquired by the broad-band seismic stations installed by INGV-OV during a temporary seismic survey (COME, COTE, COCA, and COBA) [52], and by broad-band/short-period seismic stations (EZV3, EZV4, EZV5, EZV6, EZ5, and EBMG) of the permanent Colima Telemetric Seismic Network (REd Sísmica de COlima, RESCO), managed by the Observatorio Vulcanologico de Colima of Colima University (OVC). Signals were acquired with a sampling frequency of 62.5 or 100 Hz according to the station. In Fig. 2, the map of the volcanic region with the locations of the crater area and of the seismic sensors are reported.

IV. Framework Logical Overview

Fig. 3 shows the system logical view of our designed framework along with its main modules and the relative tasks in the data flow of its pipeline. Sensing devices in the sensor network acquire the data, whose stream is collected and processed by the *Data Acquisition* Layer, transforming the collected data into a digest form for the processing. The system is designed to be easily extended with additional IoT-based devices, such as environmental sensors, even wearable ones. The *Preprocessing* step indeed is in charge of the seismic data analysis, by time sampling then storing the seismic waves collected. The *Semantic Modeling* Layer translates the processed data in Resource Description Framework (RDF) triples, according to the defined ontology schema to populate it and infers assertional data to feed and enrich the knowledge base. In the meanwhile, preprocessed data from *Data Acquisition* Layer are also given as input to the *ML-based Classification* Layer that is in charge of the seismic event detection and classification. Initially, this module works on precollected seismic data to train the ML model. Also, the knowledge base is populated adding event-related triples. Then, incoming real-time data from station sensor streams will be processed in a digested way and given as input to the neural network-based model to predict if the acquired seismic waveform is a potential seismic event. The last layer, namely, *Geophysical Data Retrieval* Layer allows querying the knowledge base by the query language SPARQL (a recursive acronym for SPARQL Protocol

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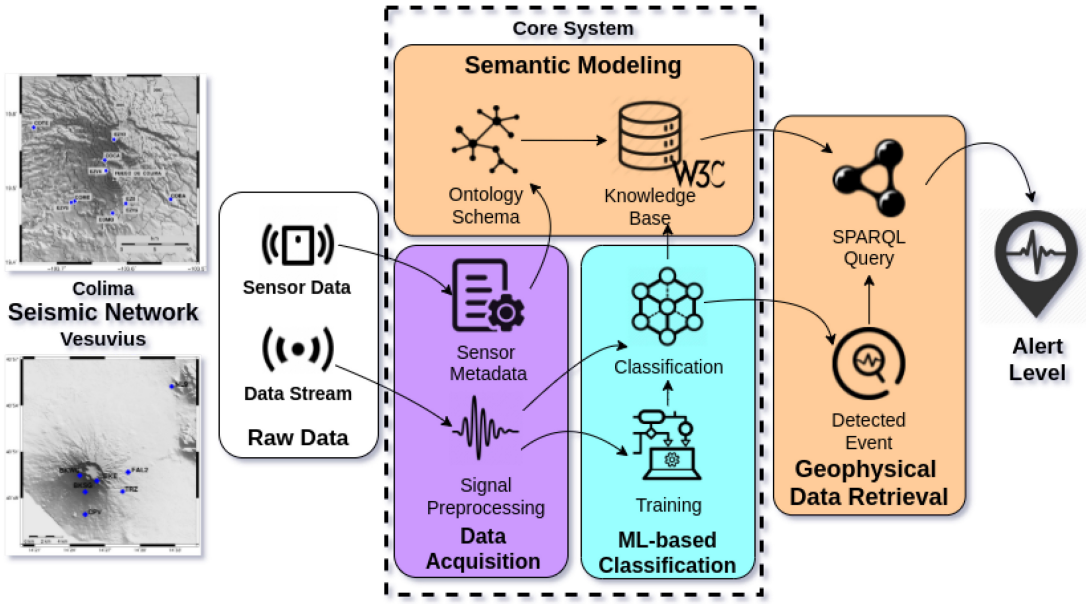


Fig. 3. Logical overview of the framework.

and RDF Query Language) that retrieves and manipulates data stored in the RDF format.

In a nutshell, our system provides a seamless infrastructure to collect data from the seismic domain, analyze them to get processable traces, and identify relevant events. The semantics plays a crucial role as an interoperability backbone to get homogeneous data representation and storage. Integrating the classification, detection, and recognition in a whole ontology-based framework can improve the capability to exploit in a unique and fruitful way all the revealed features and get an integrated view by querying the knowledge at any time.

This work focuses on presenting the core modules of the introduced framework (Fig. 3: *Data Acquisition* Layer, *Semantic Modeling* Layer, and *ML-based Classification* Layer). The complete development of the whole system is a future planned activity.

V. Data Acquisition

This module plays the role to collect and preprocess the sensing data from the sensor network. The data are mainly signals composed of one or three components. Each component corresponds to the ground velocity along a particular direction: vertical (V), north-south (NS), and east-west (EW). Seismic sensors have either one component (the vertical) like NL9 and TRZ at Vesuvius, or three components (e.g., BKE at Vesuvius and COCA, COME, COTE, and COBA at Colima). The quality of the recorded waveforms at the different stations depends on the location of the source producing the signal, so that the best signal-to-noise ratio (SNR) recordings are available for the sensors closest to the source. Thus, for each of the above-listed types of signals, we can identify a single station (called reference station) that produces the best data with highest SNR.

To summarize, the data mainly came from the following stations.

1) *VT:* BKE, BKSG, and BKWG.
2) *LP :* BKE, BKSG, and BKWG for Vesuvius; COBA, COCA, COME, COTE, EZ5, EZV3, EZV4, EZV5, and EZV6, for Colima.
3) *Blast:* CPV, TRZ, and NL9.
4) *Seismic Noise:* BKSG, BKWG, and FAL2.

These data are collected and processed by time sampling then stored in a digested way to give as input to the *Semantic Modeling* Layer as well as to the *ML-based Classification* Layer. The module is designed to process also data from IoT-based devices, such as environmental sensors, even wearable ones for the ground motion evaluation and earthquake detection from the background noise [7]. The main benefit is due to the low-cost technologies compared with the high cost of installation and maintenance of the traditional high-sensitive seismometers. The idea is to collect all sensitive data from sensing devices of different domains (seismic, sensing, geographical, etc.) by integrating heterogeneous data (IoT-based, seismologic, SAR, GPS, telemetric, tiltmetric and geodetic data, etc.) by an ontological modeling. The Data Acquisition Layer provides a common access point to collect heterogeneous data from different sources and sensing devices and then, processes them to produce a homogeneous semantic knowledge base, yet at the experimentation time, the used data were acquired, collected, and made available by the geophysical institutions (see Section III for details), while the environmental, IoT-based sensors for ground motion measurement are not yet widespread (and no data are available, to the best of our knowledge).

VI. Semantic Modeling

The *Semantic Modeling* Layer is the module in charge of generating the semantic knowledge base by exploiting the data collected by *Data Acquisition* Layer. To that purpose, an ontology schema is modeled to yield a shareable high-level

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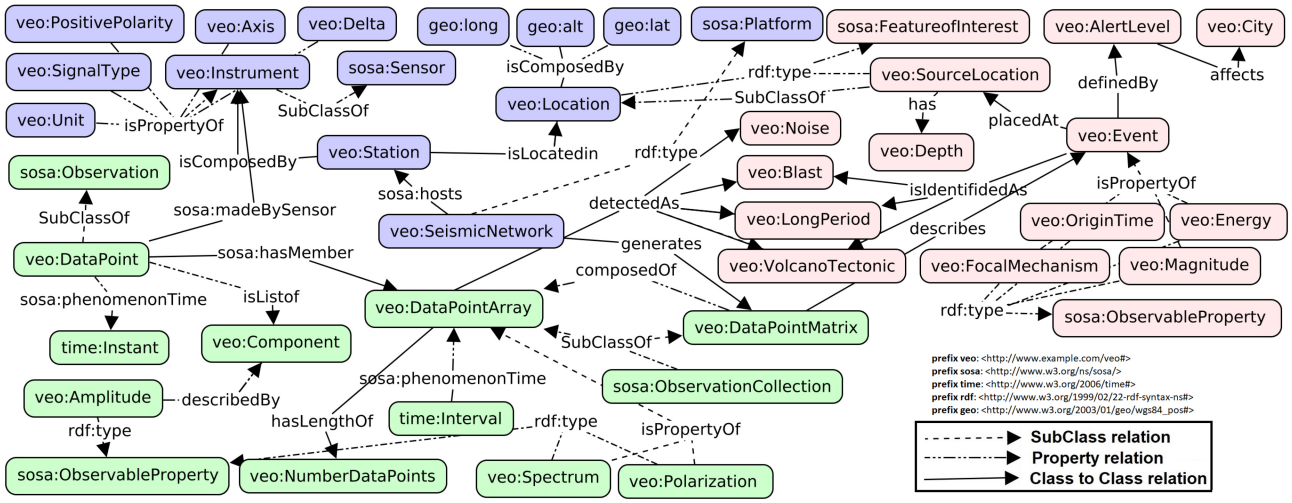


Fig. 4. Core classes of VEO.

description of the main concepts involved in the domain of interest. Indeed, the ontology model provides an integrated view of the collected data, aimed at supporting the interoperability and at facilitating the retrieval of synthesis information as a result of a homogeneous data collection. The VEO describes the main concepts from the seismic domain starting from the raw signals acquired by the sensor networks to represent and identify the seismic relevant events, such as BLs, LPs, and VTs.

The ontology model is shown in Fig. 4: concepts are in different colors: green, pink, and violet. The green concepts describe the raw data acquired by the sensors; the violet concepts are related to the seismic network description; and the pink concepts introduce the events detected. Particularly, concepts colored in violet refer to the sensor network modeling: some concepts extend known ontologies such as SSN ontology, particularly SOSA ontology. The next sections will be devoted to introduce the SSN/SOSA classes imported/extended in VEO ontology, before presenting the VEO's core classes. Moreover, some formal aspects about the composition of seismic data will be given as well.

A. SSN and SOSA Ontologies

SSN [59] is a well-known ontology model for describing sensors and their observations, the observed properties and the features of the interest domain, the deployments, and methods used for sensing [60]. At the same time, SSN provides a self-contained ontology called SOSA designed as a core vocabulary for supporting application domains and reuse-by-extension in the sensing platform environment [61]. Reasons for using SNN/SOSA rely on the high level of flexibility, and the level of standardization of that ontology. In VEO modeling, the role of SOSA/SSN concepts is crucial in the design process. Some new VEO classes extend SOSA ontology ones introducing additional functionalities. The SOSA/SSN concepts (or classes) framed in our ontology model are listed as follows.

1) *sosa:Platform:* It is the container that hosts the other entities. In our ontology model, it is specialized as a seismic network.
2) *sosa:Sensor:* It is a generic class that can represent a device, an agent (including humans), or software (simulation) for a sensing task. In our ontology model, it is a seismic domain-specific device or station for acquiring features measured by the network.
3) *sosa:Observation:* It is the class describing measurement values captured by a sensor about a property related to a Feature of Interest.
4) *sosa:ObservableProperty:* It represents an observable property or quality of the investigated phenomena. In our ontology, one of the main interesting observable properties is the amplitude (veo:Amplitude, see Fig. 4) of the seismic wave, representing a field measurement acquired by sensors, also described in Section VI-B more formally. Other properties in our ontology are calculated, such as *veo:Magnitude*, *veo:FocalMechanism*, *veo:OriginTime*, *veo:Polarization*, *veo:Spectrum*, etc.
5) *sosa:FeatureOfInterest:* This class describes the feature whose property is estimated in the course of an observation. In our VEO ontology, it can be the location of an individual sensor station (veo:Location); or it can represent the event source location where the rupture originates (veo:SourceLocation).
6) *sosa:phenomenonTime:* This class is the duration time associated with the observation reading. Depending on the Feature of Interest, it might be a time interval (time:Interval) or an instant (time:Instant), defined in the OWL Time Ontology [62].
7) *sosa:ObservationCollection:* This class has been introduced in an in-progress extension of the SSN ontology [63], that enables representing homogeneous collections of observations.

B. Seismic Wave and Event: Formal Viewpoint

The ontology model regarding the sensor data collection is given in green in Fig. 4, focusing on how the raw data coming from the station can be arranged and then processed. In particular, the main classes are the *veo:DataPoint* and the *veo:DataPointArray*. The *veo:DataPoint* represents a

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measurement from a seismic trace captured by a sensor over time. More formally, considered a sensor device s, a *DataPoint* DP_t^s at time t, can be described as follows:

$$DP_t^s = \{x_1(A,t); x_2(A,t); x_3(A,t)\} \tag{1}$$

where each component $x_i(A,t)$ with $i = 1,\ldots,3$, describes the amplitude (or, in other words, the displacement speed of the ground) of the measured wave A at time t, when the sample is taken by the sensor device s. Notice that, depending on the sensor s, DP_t^s can contain one to three components of a seismic wave (along V, NS, and EW directions of motion), which can be one to three dimensions, respectively.

A *DataPointArray* $DPA_{\Delta t}^s$ describes a sequence of *DataPoints* (DP_t^s), gathered in a time interval $\Delta t = (t_j - t_i)$ by a sensor s

$$DPA_{\Delta t}^s = \left[DP_{t_i}^s, \ldots DP_{t_j}^s\right]. \tag{2}$$

Let us notice that the time interval is undefined, when considering the whole sensor lifespan, i.e., from $t_i = 0$ (when the sensor is turn on) to $t_j = tMAX$ (the sensor device s until it is powered-off).

Once defined a sequence of the datapoints, it is possible to detect a subsequence of datapoints describing a seismic event or a generic event E.

More formally, let us consider a sensing network S composed of a set of sensor devices $S = \{s_1, s_2, \ldots, s_n\}$; an *Event* $E_{\Delta t}^S$ acquired by any sensor $s \in S$ in the time interval Δt is described as follows:

$$E_{\Delta t}^S = \left(DPA_{\Delta t}^{s_1}, DPA_{\Delta t}^{s_2}, \ldots, DPA_{\Delta t}^{s_n}\right). \tag{3}$$

The collection $(DPA_{\Delta t}^{s_1}, DPA_{\Delta t}^{s_2}, \ldots, DPA_{\Delta t}^{s_n})$ represents a generic *DataPointMatrix*, i.e., a collection of DataPointArrays gathered by each sensor of the seismic network. In particular, the DataPointMatrix can be classified as an event $E_{\Delta t}^S$ among BLs, VTs, and LPs.

C. VEO Classes

Our ontology is designed to describe the seismic domain, starting from the raw data acquired by an individual sensor or, more in general, by a sensor network, located in a volcanic environment, until to recognize the specific waveform that describes a seismic event. Moreover, the ontology models the final outcome about a possible alerting situation, supporting, for example, the authorities to make decisions in the risk scenarios.

Specifically, the class *veo:Event* describes a seismic event, detected by the analysis of the data streams collected by the seismic network, whose processing provides several information on the event, such as the source location, the origin time, moment magnitude, energy, and focal mechanism. The *veo:Event* is the main concept of the synoptic knowledge represented by the pink color in Fig. 4.

The ground shaking can be detected and identified even by processing the raw data recorded at a single station (in that case, it cannot be localized). The vector of data points (*DataPointArray*) is the trace that will be used to identify and classify the ground shaking among BLs, VTs, LPs, and the noise (that describes the continuous seismic signal produced by a variety of sources also of artificial origin).

According to the color-based ontology class partitioning, the concepts in VEO ontology are divided into *Context*, *Collection*, and *Knowledge* models describing the seismic domain from different perspectives and abstraction levels. These three schema models are described up next with the classes which form them.

1) Context Model: It describes the sensing infrastructure related to the domain of interest. The concepts, colored in violet, describe the physical layer of the sensor networks, the sensors, and complementary metadata associated with the raw data.

1) *veo:SeismicNetwork:* It extends the generic SOSA: Platform class as a specialized subclass for the seismic domain. It is an ensemble of seismic stations (e.g., TRZ and CPV), located at a certain distance to each other, according to a particular geometry. In a broader geophysical context, *Veo:SeismicNetwork* can be easily included in a general *Geophysical Network* in which all the geophysical/geochemical instruments are taken into account (e.g., tiltmeters, GPS, gravimeters, etc.).
2) *veo:Station:* It is a station composed of one or more sensors and equipped with a data logger (for local acquisition—stand-alone instrument) or with transmission apparatus (to send real-time signals to the acquisition center). Geographical coordinates are well defined for each site where the station is installed and represent a *sosa:FeatureOfInterest*. One station measures only one physical parameter and it is composed of one veo:Instrument at least.
3) *veo:Instrument:* It is a specialized class, subclass of sosa:Sensor, extended with additional properties, i.e., *veo:Unit*, which represents the measure unit used by the instrument; *veo:Delta* representing the time sampling in which a sample is taken; and *veo:SignalType* describing the signal type (analog or digital, depending on the sensor type and axis), represented by one to three components.

2) Collection Model: The classes colored in green in the VEO model are designed to support the process of collecting raw data from stations over the time. The main concepts involved in this model are listed as follows.

1) *veo:DataPoint:* It is a subclass of sosa:Observation; as stated, it represents an individual measurement of the wave signal gathered by a sensor device (*veo:Instrument*), measured by the *veo:Station*, at a given time instant *owl:Time*. A datapoint can be composed of one to three *veo:Component*s (a relation labeled *veo:asListOf* exists between *veo:DataPointArray* and *veo:Component* in the ontology model, see Fig. 4).
2) *veo:Component:* It represents the component x_i (see (1), Section VI-B) describing the ground motion along one of the directions V, NS, and EW.
3) *veo:DataPointArray:* This class extends *sosa:ObservationCollection* describing the sequence of *veo:DataPoint*s collected in a given time interval *time:Interval*, by a station sensor, according to (2).

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4) *veo:DataPointMatrix:* This class also extends *sosa:ObservationCollection*, but represents a collection of *veo:DataPointArray*s that are gathered by each sensor of the *veo:SeismicNetwork*. It can describe a *veo:Event* located in the estimated hypocenter, corresponding to the class name *veo:SourceLocation* (see (3) for details).

3) Knowledge Model: The concepts colored in pink provide a higher level description of the collected and processed data, i.e., they allow the seismic event identification and classification, accomplished by the ML-based Classification Layer. The main concepts are given as follows.

1) *veo:Location:* This class describes a geographical location, by the geographical coordinates *geo:longitude*, *geo:latitude*, and *geo:altitude*. The prefix *geo* means that the properties are imported by the WGS84 Geo Positioning Vocabulary.[1] The class *veo:Location* is related to *veo:Station* by the property *veo:isLocatedIn*, stating the geographical coordinates where the station is located; at the same time, it is a superclass of the specialized class *veo:SourceLocation* representing the hypocenter of the seismic event (i.e., the point from which the rupture originates). The source location is calculated by a suitable data triangulation through the station network.
2) *veo:Event:* It is the class describing the source event detected. The instances of this class are indeed the Earth ground motions detected by the sensor network and processed to identify the *veo:SourceLocation*, at *veo:OriginTime* (representing the starting time when the rupture occurs at the hypocenter). In our context and experimentation, the classes *veo:Blast*, *veo:VolcanoTectonic*, and *veo:LongPeriod* are the possible events that the ML-based classification Layer identifies. The *veo:Event* class has several properties (sosa:ObservableProperty), e.g., *veo:Magnitude*, *veo:FocalMechanism*, *veo:OriginTime*, and *veo:Energy* which can be appropriately associated with a particular event.
3) *veo:Blast:* It is the class that describes the blasts (such as the quarry blasts occurred in Nola or Terzigno caves or the sea explosions due to illegal fishing).
4) *veo:VolcanoTectonic:* It is the class describing VT earthquakes.
5) *veo:LongPeriod:* It is a class describing LP events.
6) *veo:Noise:* Is the class describing the background seismic signal.
7) *veo:City:* It is a class describing the city involved in a seismic event. It is related to the *veo: AlertLevel* class, which describes the impact (i.e., consequences) in terms of alert level the event could have on the cities.
8) *veo:AlertLevel:* It is a class that provides the alerting level, evaluated after the event detection and the estimation of its severity.

Let us notice that the classes *veo:Blast*, *veo:VolcanoTectonic*, *veo:LongPeriod*, and *veo:Noise* are populated by the ML-based Classification Layer: whenever a new ground shaking is detected by a sensor, a new statement is added to the knowledge base to assert that signal is recognized and classified as a BL, VT, LP, or commonly noise.

[1]Available: http://www.w3.org/2003/01/geo/wgs84_pos.

VII. ML-Based Classification: Analysis and Experimental Results

The *ML-based Classification* Layer is in charge of classifying preprocessed raw seismic data and then predict the type of events, especially when they can be critical. Predicted events feed the knowledge base by enhancing the population of classes, such as *veo:Blast*, *veo:VolcanoTecnotic*, and *veo:LongPeriod* and also *veo:Noise* (Fig. 4) in our domain ontology VEO. A comparative study has been achieved by carrying out two ML classification methods: 1) MLP and 2) CNN.

A. Data Description

The dataset was built by collecting and processing seismic signals from Vesuvius and Colima volcanoes. It consists of 4008 signals in the SAC format (Seismic Analysis Code).

Due to the different types of configurations and samples taken from the stations, all the raw samples were resampled at 100 Hz ($\Delta t = 0.1$ s) in the *Data Acquisition* Layer. This feature guarantees the consistent data translation by the Context Model, and also the sample normalization to get a uniform 2400 *veo:DataPoint* signal, equivalent to 24 s. Indeed, 2000 samples correspond to raw event detection by the sensor, and the additional 400 samples are taken for signal isolation to prevent overlapping.

Using the previous method, the collected seismic data from Vesuvius consist of 416 VTs, 464 BLs (quarry and sea explosions), and 363 LPs. The data from Colima are 2765 LPs signals. In addition to these event types, data from *the continuous seismic background noise* are picked from the Vesuvius stations. To have a balanced signal among all the wave types, samples of noise were taken by selecting the same time window 24 s long, to extract each *veo:DataPointArray*, thus archiving over 28 000 samples for this type.

To fix the unbalancing classes issue, the data augmentation was accomplished by inverting the polarity of each original seismic trace, doubling the dataset size for each event class (excluding the noise background signals).

B. ML-Based Model Design

The first ML method is MLP backpropagation algorithm. It consists of three types of layers: the input layer, output layer, and hidden layer with 42 neurons, with *Relu* activation function. The neurons in the MLP are trained with the backpropagation learning algorithm for updating the weights, calculated by descending gradient (*adam*). The training parameter configuration is given as follows.

1) *Hidden Layers:* 42.
2) *Batch Size:* 50.
3) *Training Rate:* 0.001.
4) *Gradient influence:* 0.9.
5) *Max. Iterations:* 640.

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TABLE I
SIGNAL CLASSIFICATION—EXPERIMENTAL DESIGN: BL:BLAST, VT: VT EARTHQUAKE, LP: LONG-PERIOD, LPV: LONG PERIOD FROM VESUVIUS, LPC: LONG PERIOD FROM COLIMA, NO: NOISE, PRECISION (P), RECALL (R), AND F1-SCORE (F1)

		Event Class					MLP			CNN		
Exp.	Classes	BL	VT	LPV	LPC	NO	P	R	F1	P	R	F1
1	2	–	–	–	5530	5530	0.98	0.98	0.98	1.00	0.98	0.98
2	2	–	–	726 + 5530		6256	0.98	0.98	0.98	0.99	0.99	0.99
3	2	832 + 928 + 0 + 900				2765	0.99	0.98	0.99	1.00	1.00	1.00
4	4	416	464	–	450	450	0.94	0.94	0.94	0.93	0.93	0.93
5	4	832	928	–	900	900	0.97	0.98	0.97	0.98	0.98	0.98
6	5	832	928	726	1000	1000	0.93	0.93	0.93	0.97	0.98	0.98
7	3	–	–	726	1000	1000	0.94	0.94	0.94	0.98	0.99	0.98
8	3	–	928 + 726		2765	2765	0.94	0.95	0.94	0.98	0.98	0.98

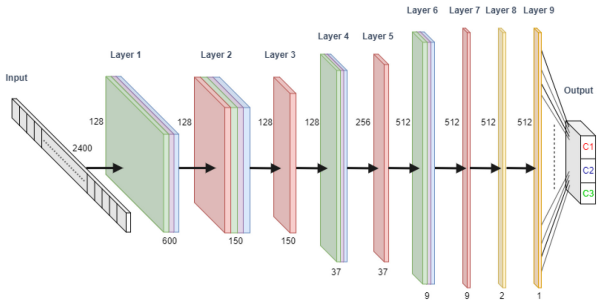


Fig. 5. Model of layers for CNN with the layer colors. Green: Convolution1D, purple: batch normalization, blue: activation (ReLu), red: MaxPooling1D, yellow: lambda (mean), and orange: dense (SofMax).

The second ML method is based on a CNN architecture consisting of nine blocks (as shown in Fig. 5). *Layer 1* is composed of a convolutional 1 dimension, with 128 filters used for our time-series data of size 2400, a kernel size of 80, and a stride of 4. Then, a batch normalization layer followed by a ReLU activation layer. *Layer 2* is composed of a max-pooling layer with size 4, while the remainder has the same structure of the first block: CNN-1D with 128 filters and kernel size and stride equal to 3 and 1, respectively. Then, *layer 3* consists of a second max-pooling layer of size 4 that feds *layer 4* similar to the first one: a CNN1D with 256 filters, kernel size equals to 3 and stride equal to 1. Then, *layer 5* has another max-pooling layer with the same structure of the previous one; *layer 6* has the same structure of the fourth block except than the number of filters equals to 521 in the CNN-1d layer. The final blocks, namely, *layers 7–9* are, respectively, a max-pooling layer with a size of 4, the Lambda "mean" layer and, finally, a Dense layer with softmax activation, transforming 512 tensors into one dimension, to get the final class distribution of input data. The configuration setting for training is given as follows.

1) *Batch Size:* 128.
2) *Epochs:* 200.
3) *Optimizer:* Adam.
4) *Activation Function:* ReduceLROnPlateau.
   a) *Monitor:* "val_loss";
   b) *Reduction Factor:* 0.5;
   c) *Learning Rate:* 0.0001.

C. Experiment Settings

Raw data collected by the seismic monitoring network of Vesuvius and Colima are analog signals, telemetered by radio links or by dedicated line to the station network system. A preliminary preprocessing phase was achieved by exploiting a widely used method for extracting spectral features in signal processing: the linear predictive coding (LPC). It is used with the purpose to represent the spectral envelope of a numeric signal in a compressed form, using the information from a linear predictive model [31]. A signal parametrization in the time domain allows extracting information about the waveform. The extracted features from the input of the *ML-based Classification* module.

The neural network models were developed in Python, with TensorFlow, and SkLearn ML frameworks, the pysac library was used for the handle of raw seismic files, Owlready2 and EDFlib libraries as driver for the ontology connection. Finally, the SPARQL language was employed for querying the triple store knowledge base. The experiments were run on a processing Ubuntu 20.04LTS computing node equipped with 8 Xeon 5220@2.2-GHz processor and 32 GB of RAM.

Regarding the data management, the training data splitting policy was 70:20:10, leaving, 70% for the training set, 20% for the validation set, and 10% for the test set. The training parameter configurations are reported in Section VII-B and are used in all the experiments.

Table I shows the dataset configurations used for each experiment and the event classes involved in the classification. The performance for MLP and CNN is given in terms of the macro-average for the precision (P), recall (R), and F1-score (F1). The Precision (P) is the ratio between the true positives (tps) and all the positives (tp + fp) $P = [\text{tp}/(\text{tp} + \text{fp})]$. The Recall (R) measures how our model correctly identifies tps. It is the ratio between the tps and the sum of tps and those one that the model does not recognize as positive (fn) $R = [\text{tp}/(\text{tp} + \text{fn})]$, and finally, the F1-score (F1) is defined as the harmonic mean of P and R and it measures the accuracy of our models on the datasets: $F1 = [(2 \times \text{P} \times \text{R})/(\text{P} + \text{R})]$.

D. Results

Experiments highlight the accuracy of the proposed methods in the seismic event classification, including BLs, VTs, and LPs; at the same time, interesting insights in the seismic wave learning from the event combination are revealed by the classification, even considering the background noise data in the discrimination of effective seismic events.

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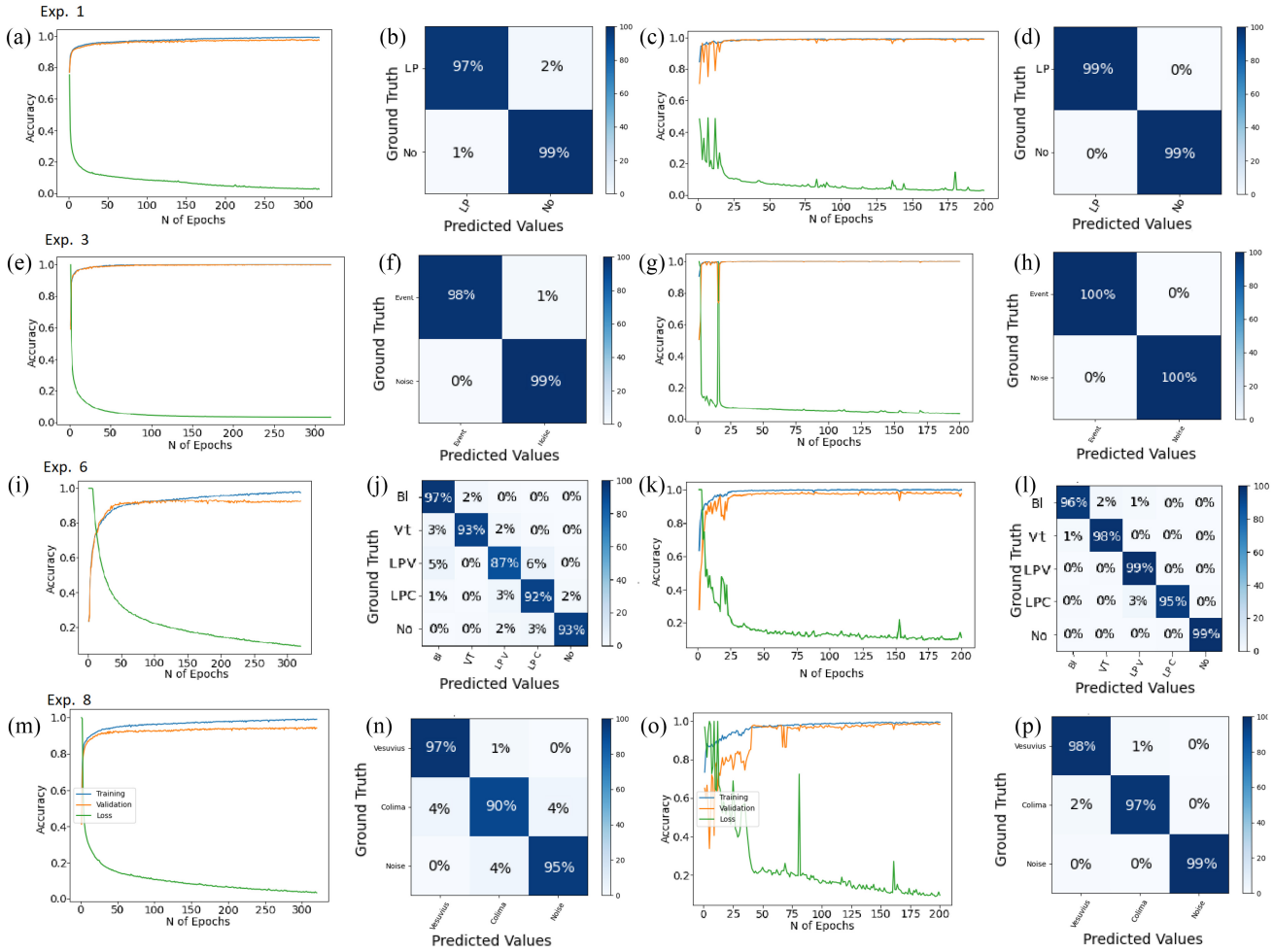


Fig. 6. Performance results of MLP and CNN for experiments #1, #3, #6, and #8. For each experiment, the chart of training and validation accuracy, and training loss (a), (e), (i), and (m) as well the confusion matrix (b), (f), (j), and (n) are showed for MLP; similarly, the chart of training and validation accuracy, and training loss (c), (g), (k), and (o) as well the confusion matrix (d), (h), (l), and (p) are showed for CNN.

Table I reports eight experiments, some of them are furthermore detailed providing training and validation tendencies along with the confusion matrices in Fig. 6.

Experiments #1–#3 propose a binary classification, grouping the seismic samples in different ways: by types and also by sources. The goal is to assess if the classifier is able to discriminate the seismic wave from the background noise (the other class is composed of all noise data).

Experiment #1 considers a dataset composed of 5530 LPs traces from Colima and 5530 traces of noise data. Since the size of Colima LPs traces is not big as for noise, a data augmentation process has been applied inverting the polarity of the Colima LP traces. The experimental results show an accurate classification with respect to the two designed classes, for both the classifiers. Fig. 6(a)–(d) shows the training and validation accuracy as well as the training loss, over the number of epochs. Moreover, the confusion matrices are also provided. Notice that LP recognition is very relevant for volcano hazards because the occurrence of LPs is very often associated with a renewal of volcanic activity.

Experiment #2 considers the same two classes but the dataset is larger: now the LP class includes the LPs from both volcanoes. As already said, merging these two classes makes sense because the LPs are similar in waveform, duration, and spectral content. The class of noise samples increases accordingly. Performance results are comparable with the previous experiment, showing that the models are able to discriminate the LPs from noise.

Experiment #3, reported also in Fig. 6(e)–(h), proposes a binary classification targeted at the discriminating *event* from *no event*. The models were trained considering two classes: one composed of all the seismic events: BLs, VTs, and LPs from both Vesuvius and Colima (*event* class), whereas the other class includes the noise (*no event* class). The two classes have the same sample size. The performance shown in Table I emphasizes that both the network models can discriminate input data correctly.

In Experiment #4, the classification aims at detecting the different seismic events. The dataset is now composed of four classes: 1) BLs; 2) VTs; 3) LPs from Colima; and 4) noise data. To ensure balanced classes, each class is composed of about 400 samples. No data augmentation is applied in this experiment. The results are good even if the values for P, R, and F1 are slightly lower compared to the

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previous experiments. It is due to the increase in the number of classes.

Experiment #5 is similar to the previous one, but the sizes of the event classes are increased. Precisely, samples for the blasts and VTs are obtained by data augmentation. Sizes for BL and VT classes are 832 and 928, respectively, i.e., they are duplicated. Sample sets with similar sizes are selected for building LPC (LP from Colima) and NO (noise) classes guaranteeing a homogeneous class balancing. Performance results for both the experiments are satisfying with a slight improvement with the increased size of datasets.

Experiment #6, reported in Fig. 6(i)–(l), includes all the individual classes: BL, VT, and LP from Vesuvius have duplicated size, whereas the size of LP from Colima and noise (NO) is restricted to consider just 1000 traces for class balancing purposes. Performance results for MLP in this experiment are slightly lower than the other experiments, probably due to the increased number of classes.

This case represents a strong improvement in the detection process of volcano observatories, because it allows the prompt distinction among those events having a volcanic origin from those not volcanic (such as BLs).

Experiment #7 aims at assessing the accuracy of the two models in discriminating LP from Colima, LP from Vesuvius, and noise. By balancing the class size (augmenting LP samples from Vesuvius), the CNN presents more accurate data classification. Although the similarity of LP waveforms, the model recognizes even slight differences between them, distinguishing LPs acquired at Vesuvius from those at Colima. Let us remind that LPs are a marker of the renewal of the volcanic activity.

Finally, the last experiment (Experiment # 8) aims at recognizing the "signature" of the volcano, taking into account the combined acquisition of all VTs (from Vesuvius) and LPs (from both Vesuvius and Colima). Therefore, the classes of this experiment are Vesuvius, Colima, and Noise.

This case could be a step toward a very global database, where seismic signals from volcanoes worldwide are recognizable. The results reported in Fig. 6(m)–(p) show the trends for training, validation, and loss along with the confusion matrix for both models. Also in this experiment, the performance of the two models is very good, especially for the CNN that gets 0.98 for P, R, and F1-score.

The described experiments show the effectiveness of our models in classifying seismic signals, proposing different classification, according to the event classes taken into account. Similar experimentation is described in [64]: two classification models, namely, Random Forest and SVM are employed to classify 109 609 seismic events at Ubinas Volcano, Peru, exploiting 102 features gathered from both acoustic and seismic fields. The approach uses a rich feature space compared to ours, with the consequent costly and time-consuming feature extraction activity and, as stated by authors, the difficulty of finding features that effectively separate the observations into their classes. Performance in event recognition seems comparable to ours, especially for Experiment #6 that considers five classes (versus the six of the approach), including seismic and not seismic events. Our model has 93% F1-score versus $93.5\% \pm 0.50\%$ accuracy of [64]. Many works in [65] present seismic event classification approaches, ranging from an SVM model [35] to RNNs [29] and also HMMs [30], trained on one [31] to more types of features [64], with several seismic event classes [64]–[66].

In particular, in [67], some classifiers have been trained considering one and multichannel sensors to model the behavior of the volcano. An MLP model has been employed even though the best performance is associated with the SVM model. The best results are obtained when the models are enhanced with the empirical mode decomposition (EMD), an adaptive decomposition method for nonstationary signals, such as the seismic events. In [66], four volcano signals, namely, VT, LP, Tremor (TR), and Tectonic (TC) collected from Llaima Volcano in Chile were analyzed using different approaches: 1) raw data; 2) spectrograms; and 3) wavelets. The approach presents several experiments whose classification performances have been assessed by using 2-D CNNs (used to classify 2-D-spectrogram images of seismic events), and a long short-term memory (LSTM) and MLP used as a baseline. The result for 2-D-CNN exploiting continuous wavelet transform (CWT, used to decompose a signal into wavelets and considered strongly robust to the presence of outliers and noise), is 94.66% F1-score, averagely. CWT input data followed by an MLP architecture show similar performance (F1-score is 95.47% averagely), confirming the effectiveness of this simple model. Finally, the authors propose a novel DNN called SeismicNet. It produces classification F1-score values equal to 91.75% and 93.02% applied on the raw seismic signals and on their wavelet samples, respectively. Our performances for both MLP and CNN models are always higher than 90% F1-score, evaluated considering the raw seismic signals.

VIII. Conclusion

This work introduces a semantically-enhanced framework for seismic event detection, starting from data acquired by two seismic sensor networks. An *ad hoc* defined ontology describes the seismic domain, modeling seismic signals for the seismic event recognition. It has been built on the well-known SSN/SOSA ontology, modeled to describe sensor-based systems. Besides the ontology modeling, the framework accomplishes classification/detection of seismic events to feed the knowledge base of new triple-based assertions. Several experiments were carried out on seismic signals recorded at two different volcanoes: 1) Mt. Vesuvius (Naples, Italy) and 2) Colima volcano (Mexico), achieved by means of comparative analysis of two ML-based methods, namely, MLP and CNN models. The obtained results highlight the effectiveness of the proposed framework in terms of flexibility and high accuracy. Successful detection and classification are crucial for evidencing any alteration in the volcano dynamics; our framework is designed indeed to support human experts, providing comprehensive access to the knowledge base for a prompt alert in cases of volcanic crises. The synergy between Artificial Intelligence techniques and the SWTs provides a solid and formal infrastructure to annotate discovered seismic patterns, by the semantic integration of raw data of different

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types and coming from heterogeneous sources. Besides the seismic networks with highly sensitive and costly seismometers, wearable, low-cost IoT sensors have also been developed for the earthquake detection.

Our approach provides a forefront framework designed to be easily extended in a larger sensing domain, including not only traditional station network approaches but also IoT-based applications and devices, to build a comprehensive knowledge base to support users in seismic alerting identification.

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